



“ML BASED BCI WHEEL CHAIR SYSTEM”

A PROJECT REPORT

SUBMITTED by

MOHAMMED JAWAD PATEL (190303107061)

RUDRESH PATEL (190303107062)

REHAN KHAN (210303107901)

KUSHANK SINGH (190303107083)

In fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

**ELECTRONICS AND COMMUNICATION ENGINEERING
DEPARTMENT**

Under the guidance of

Dr. Ramji Gupta



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
PARUL INSTITUTE OF ENGINEERING AND TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY**

**PARUL UNIVERSITY, VADODARA
March 2023**

CERTIFICATE

This is to certify that the project work entitled “**ML BASED BCI WHEEL CHAIR SYSTEM**” for subject Project -II (203107451) of 8th Semester, Group No. PU_ECE09 has been successfully completed by **MOHAMMED JAWAD PATEL (190303107061), RUDRESH UMESHBHAI PATEL (190303107062), KUSHANK SINGH (190303107083), REHAN KHAN (210303107901)**, fulfilment of final year of the Bachelor of Technology (B.Tech) in Electronics and Communication of Parul University in Academic year 2022-2023 is a record of the group of above student’s own work carried out by them under my supervision and guidance. Information derived from the published work of others has been acknowledge in the text and a list of references is given.



Year: 2022-23

Supervised by

Dr. Ramji Gupta

ELECTRONICS AND COMMUNICATION ENGINEERING DEPARTMENT

PARUL INSTITUTE OF ENGINEERING AND TECHNOLOGY

PARUL UNIVERSITY

Dr. Ramji Gupta

(Project Supervisor)

Prof. Anuradha Gcharge

(Project Co-ordinator)

External Examiner

Dr. Kalpesh Jadav
(HOD)

ABSTRACT

This report presents machine learning based SSVEP BCI wheelchair that incorporates frequency target stimuli as a means to direct the intention of a user to go in different direction. Traditional BCI system has a limited accuracy, more prone to noise, requires extensive training. To overcome these limitations this report presents SSVEP (Steady State Visual Evoked Potential) based BCI system to control a wheelchair in forward, left, right and in stop position. This dataset gathered SSVEP-BCI recordings of 35 healthy subjects (17 females, aged 17-34 years, mean age: 22 years) focusing on characters flickering at different frequencies (8-15.8 Hz). 4 SSVEP frequency were elicited as targeted stimuli. The EEG signals were acquired from occipital region and features were extracted using FFT (Fast Fourier Transform). The machine learning algorithm named SVM (Support Vector Machine) was used to perform the decoding of SSVEP feature vectors.

ACKNOWLEDGEMENT

We are deeply indebted to our Internal Guide **Dr. Ramji Gupta** for his constant guidance, motivation, suggestion and encouragement through our literature review and project work and for being helpful in making our presentation more effective. He has devoted significant amount of his valuable time to plan and discuss the project work. Without his experience and insights, it would have been very difficult to do quality work.

We are all thankful to our project Coordinator **Prof. Anuradha Gharge** of Electronics & Communication Engineering Department, Parul Institute of Engineering and Technology, Parul University for directing us to carry out our project at given time and by discussing about reading, searching and writing of literature review and project work for generous assistance and cooperation.

We would like to express our sincere gratitude to **Dr. Kalpesh Jadav** of E&C, HOD, PIET, Parul University for helping us in this project work.

Last but not the least, we would like to thank our college, faculties of ECE department, friends for helping out in difficult situations during my literature review preparation and also our parents for supporting us throughout the literature review and project work.

Above all we would like to thank almighty for giving us strength to do the literature review and the project work.

MOHAMMED JAWAD PATEL (190303107061)

RUDRESH PATEL (190303107062)

REHAN KHAN (210303107901)

KUSHANK SINGH (190303107083)

TABLE OF CONTENTS

Certificate.....	i
Abstract.....	ii
Acknowledgement.....	iii
Table of Contents.....	iv
List Of Figures.....	v
Plagiarism Report.....	iv
 Chapter 1: Introduction.....	 1
1.1 Introduction.....	
1.2 Justification of Problem.....	
1.3 Scope.....	
1.4 Aim.....	
1.5 Objective.....	
Chapter 2: Literature Review.....	3
2.1 Literature Review.....	
Chapter 3: Experimental Setup and Methodology.....	11
3.1 Experimental Setup.....	
3.2 System Overview.....	
3.3 Methodology.....	
3.4 Software Description.....	
3.5 Working of The Project.....	
Chapter 4: Results.....	17
4.1 Results & Stepwise Working.....	
4.2 Observation.....	
Chapter 5: Conclusion & Future Perspective.....	22
5.1 Conclusion.....	
5.2 Future Perspective.....	
References.....	23

LIST OF FIGURES

Figure 3.3.1 Flowchart of Process.....	14
Figure 4.1.1 Removing DC Offset from The Train Data.....	17
Figure 4.1.2 Removing Excess Frequency & Computing FFT From the Train Data.....	17
Figure 4.1.3 Removing DC Offset from the Test Data.....	18
Figure 4.1.4 Removing Excess Frequency & Computing FFT From the Test Data.....	18
Figure 4.1.5 Computing the 1 st And 2 nd Harmonics from the EEG Data.....	19
Figure 4.1.6 Using SVM To Compute the Relation Between Feature Matrix & Frequency Labels	19
Figure 4.2.1 Raw EEG Signal & DC Corrected EEG Signal.....	20
Figure 4.2.2 Unfiltered EEG Signal & Bandpass Filtered EEG Signal	20
Figure 4.2.3 FFT of EEG Signal Recorded.....	21
Figure 4.2.4 FFT of EEG Signal Recorded.....	21

PLAGUARISM REPORT

Originality report

COURSE NAME

Final year project

STUDENT NAME

RUDRESH PATEL

FILE NAME

RUDRESH PATEL - B.tech Final year Project Gropu 11

REPORT CREATED

Mar 5, 2023

Summary

Flagged passages	7	4%
Cited/quoted passages	3	3%
Web matches		
neurotechx.com	5	3%
ijsce.org	2	2%
mdpi.com	2	1%
researchgate.net	1	0,5%

1 of 10 passages

Student passage **FLAGGED**

This report presents SSVEP (Steady State Visual Evoked Potential) based BCI system to control a wheelchair in forward, left, right and in stop

Top web match

Abstract– **This paper presents a Steady State Visual Evoked Potential (SSVEP) based Brain Computer Interface (BCI) system to control a wheelchair in forward, backward, left, right and in stop...**

BCI Based Wheelchair Control using Steady State Visual Evoked ... <https://www.ijsce.org/wp-content/uploads/papers/v3i3/C1621073313.pdf>

2 of 10 passages

Student passage **CITED**

This dataset gathered SSVEP-BCI recordings of 35 healthy subjects (17 females, aged 17-34 years, mean age: 22 years) focusing on characters flickering at different frequencies (8-15.8 Hz).

Top web match

This dataset gathered SSVEP-BCI recordings of 35 healthy subjects (17 females, aged 17-34 years, mean age: 22 years) focusing on 40 characters flickering at different frequencies (8-15.8 Hz with an...

moabb.datasets.Wang2016 -

NeuroTechX <http://moabb.neurotechx.com/docs/generated/moabb.datasets.Wang2016.html>

3 of 10 passages

Student passage FLAGGED

...therefore one of the mobility restoration options for people with movement disorders such as amyotrophic lateral sclerosis (ALS), brainstem stroke, brain or spinal cord injury

[Top web match](#)

It has mainly targeted patients with neurological diseases such as amyotrophic lateral sclerosis (ALS), brainstem stroke, multiple sclerosis, and high spinal cord injury.

Motor-Imagery EEG-Based BCIs in Wheelchair Movement and Control <https://www.mdpi.com/1424-8220/21/18/6285>

4 of 10 passages

Student passage FLAGGED

These research papers have been carefully analysed, and the main characteristics have been mentioned in terms of feature extraction, pre-processing, signal acquisition, and classification methods...

[Top web match](#)

In this section, several existing applications of MI EEG-based BCIs for wheelchair movement and control are illustrated. These studies are carefully analyzed, and the main characteristics in terms of...

Motor-Imagery EEG-Based BCIs in Wheelchair Movement and Control <https://www.mdpi.com/1424-8220/21/18/6285>

5 of 10 passages

Student passage FLAGGED

...signals will be recorded using a 64-channel EEG device. The reference electrode was placed on the right earlobe and the electrodes were arranged according to the international 10-20 system

[Top web match](#)

measure the reference value for a EEG electrode cap and EEG signals before measuring EEG, the reference electrode was attached on the earlobe like a clip, and according to the International 10-20...

Electrode cap arrangement diagram and brain lobe. -

ResearchGate https://www.researchgate.net/figure/Electrode-cap-arrangement-diagram-and-brain-lobe_fig2_333219800

6 of 10 passages

Student passage CITED

- Continuous EEG data was segmented into 6 s periods (500 ms before stimulation and 5.5 s after the start of stimulation). Then reduce the period...

[Top web match](#)

The **continuous EEG data was segmented into 6 s epochs (500 ms pre-stimulus, 5.5 s post-stimulus onset)**. The epochs were subsequently downsampled to 250 Hz. Thus each trial consisted of 1500 time...

moabb.datasets.Wang2016 -

NeuroTechX <http://moabb.neurotechx.com/docs/generated/moabb.datasets.Wang2016.html>

7 of 10 passages

Student passage FLAGGED

...of stimulation). Then reduce the period to 250 Hz. **Finally**, this data is stored as double-precision floating-point values in MATLAB and labelled as subject

[Top web match](#)

Thus each trial consisted of 1500 time points. **Finally**, these data were stored as double-precision floating-point values in MATLAB and were named as subject indices (i.e., S01. mat, ..., S35.

moabb.datasets.Wang2016 -

NeuroTechX <http://moabb.neurotechx.com/docs/generated/moabb.datasets.Wang2016.html>

8 of 10 passages

Student passage CITED

...floating-point values in MATLAB and labelled as subject indices. **For each file, the data loaded into MATLAB results in a 4-dimensional matrix called "data" with dimensions (64, 1500, 40, 6) [26].**

[Top web match](#)

mat). **For each file, the data loaded in MATLAB generate a 4-D matrix named 'data' with dimensions of (64, 1500, 40, 6).**

moabb.datasets.Wang2016 -

NeuroTechX <http://moabb.neurotechx.com/docs/generated/moabb.datasets.Wang2016.html>

9 of 10 passages

Student passage FLAGGED

the FFTs of all connected channels, the data from Oz&T10 was chosen for further system development, as the strongest SSVEP was observed at Oz. The coefficients of the fundamental and second harmonics...

[Top web match](#)

From **the FFT of all the connected channels, the data from Oz —A2 was selected for further system development as strongest...The coefficients of the fundamental and second harmonics of all the four...**

BCI Based Wheelchair Control using Steady State Visual Evoked ... <https://www.ijscce.org/wp-content/uploads/papers/v3i3/C1621073313.pdf>

10 of 10 passages

Student passage FLAGGED

...on SVM, goes through several stages. First, data from **35 healthy** people who focused on **40 characters flickering at different frequencies** was gathered. Six blocks, each having 40 randomised trials...

[Top web match](#)

This dataset gathered SSVEP-BCI recordings of **35 healthy** subjects (17 females, aged 17-34 years, mean age: 22 years) focusing on **40 characters flickering at different frequencies** (8-15.8 Hz with an...

moabb.datasets.Wang2016 -

NeuroTechX <http://moabb.neurotechx.com/docs/generated/moabb.datasets.Wang2016.html>

CHAPTER 1

INTRODUCTION

1.1 Introduction

Severe motor disabilities present a variety of difficulties for everyday living, including mobility and independence restrictions. Conventional mobility devices, such as wheelchairs, necessitate user movement or input, which for those with severe motor limitations can be challenging or impossible. Using brain impulses instead of movement or input, BCI technology enables users to control external equipment. Using steady-state visually evoked potentials (SSVEPs), which are brain impulses elicited by straightforward visual stimuli that flicker at particular frequencies, is one viable method of BCI technology.

In this study, we design SSVEP based BCI Wheel chair system that can decode user's intended direction using SVM algorithm. The user's SSVEP responses are elicited by the system using visual stimuli, and these responses are then recognized and categorized using machine learning algorithms. The system's effectiveness in regulating the wheelchair's movements has been evaluated on a group of healthy individuals, and the findings bear this up.

1.2 Justification of Problem.

For those with severe motor limitations, traditional brain-computer interface (BCI) wheelchair systems have a difficult time offering dependable and efficient mobility options. These systems use electroencephalography (EEG) signals, which are susceptible to noise and interference from other sources, to determine the user's intended direction of movement. Furthermore, user training and calibration for classic BCI systems are frequently time-consuming and stressful for the user.

We demonstrate a BCI wheelchair system based on machine learning that can identify the user's intended direction of movement using steady-state visually evoked potentials (SSVEPs). Compared to conventional EEG signals, SSVEPs are less prone to noise and interference, and they can be triggered by employing straightforward visual stimuli. The amount of training and calibration time needed is greatly decreased by using a dataset of SSVEP signals that were acquired from the user during an initial calibration phase to train the machine learning algorithms.

1.3 Scope

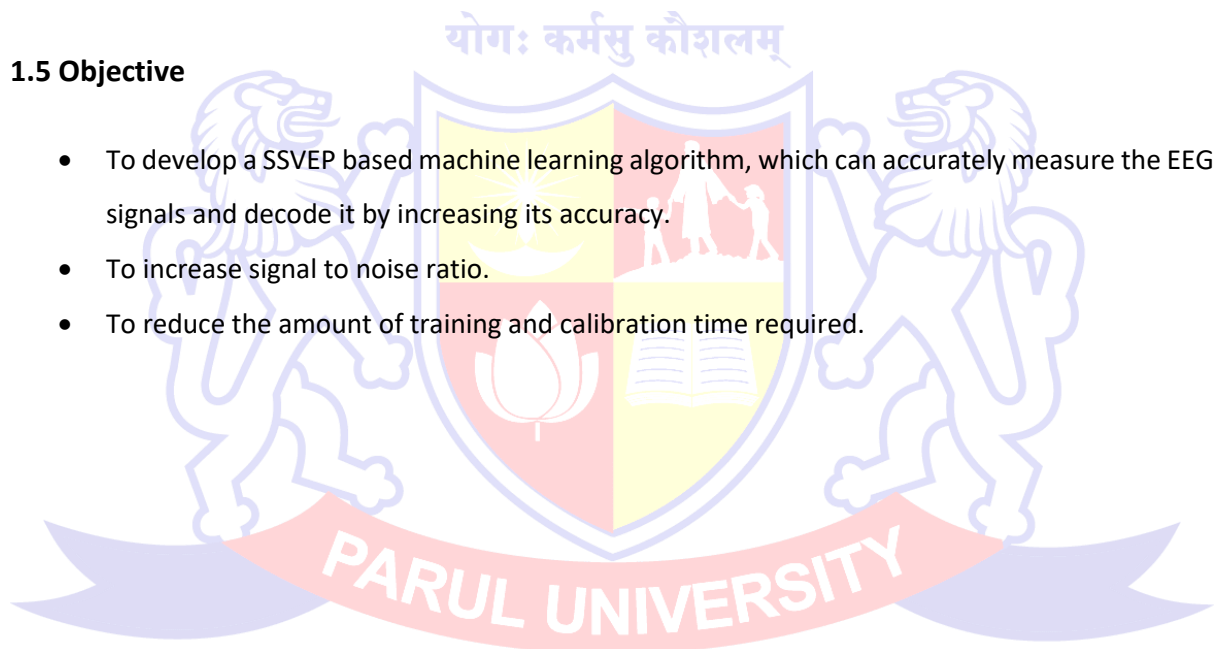
The development of brain-controlled wheelchairs involves the installation of multiple hardware and software processes. It brings out a key challenge faced by all motor-disabled patients & it will be helpful in easing their life in many ways. And the scope is quite good as 75 million people need wheelchairs on a daily basis which represents 1% of the total world's population.

1.4 Aim

The aim of this project is to develop and test a BCI machine learning-based SSVEP wheelchair algorithm that provides different directions of movement for individuals with severe motor disabilities.

1.5 Objective

- To develop a SSVEP based machine learning algorithm, which can accurately measure the EEG signals and decode it by increasing its accuracy.
- To increase signal to noise ratio.
- To reduce the amount of training and calibration time required.



CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE REVIEW:

Millions of people with disabilities around the world suffer from reduced mobility and rely on powered wheelchairs to carry out their daily activities and a lot of research has been done for this purpose by using brain signals to directly control the wheelchair. BCI is therefore one of the mobility restoration options for people with movement disorders such as amyotrophic lateral sclerosis (ALS), brainstem stroke, brain or spinal cord injury, cerebral palsy, muscular dystrophy, and multiple sclerosis.

In this portion, we have illustrated various existing applications of EEG-based BCI for wheelchair movement and control. These research papers have been carefully analysed, and the main characteristics have been mentioned in terms of feature extraction, pre-processing, signal acquisition, and classification methods along with wheelchair performances evaluation.

1. Tsui, C.S.L.; Gan, J. Hu, H. on A Self Paced Motor-Imagery Based Brain Computer Interface used for control in Robotic Wheelchair Clin . EEG Neurosci. 2011, 42, 225–229 [23]. Tsui presented a simple two-class MI-based BCI for self-stimulation for wheelchair control. This system allowed the user to plan the route and have full control over the wheelchair. Based on the laser range finder, an automatic obstacle avoidance system is integrated into the control mechanism of the robotic truck. After practicing on the simulator, the system was tested online in the Robotics Arena at the University of Essex and the experiment was conducted with two subjects.
2. Carrino, F.; Dumoulin, J.; Mugellini, E.; Khaled, O.A.; Ingold, R. A self paced BCI system used for controlling an electric wheelchair: Evaluation of a commercial, low-cost EEG device. ISSNIP Bio-signals and Bio-robotics Conference: Bio-signals and Robotics for Better and Safer Living (BRC) [22] in Rio de Janeiro, Brazil. Carrino in the month of February 2012 proposed a user-friendly BCI automated driving system that enables users to drive a powered wheelchair using a commercially available EEG headset and motor imaging approach. Inexpensive EEG devices have yielded interesting results, but the authors said they are largely unusable for self-regulating systems in error-sensitive situations. A problem has occurred. More specifically,

failures in the classification process resulted in unpredictable wheelchair behaviors and strong impairments in users focused on motor imagery tasks. Finally, since this problem leaves no navigation for untrained subjects, the test was conducted with a single subject and used real-world gestures that are less susceptible to affective disturbances.

3. Choi, K on control of vehicle in real time based on EEG signals and system evaluation. Graefe Arch. Clin. Exp. Ophthalmol. 2011, [24] A new wheelchair system controlled by EEG signals was developed by Choi et al. Built using effective signal processing methods, it allows paralyzed people from the neck down to interact more freely with society. The implemented system was evaluated through bar control and obstacle avoidance experiments with three subjects. The authors confirmed that the proposed wheelchair system performs almost the same as a joystick-controlled wheelchair. सु कौशलम्
4. Li, J.; Liang, J.; Zhao, Q.; Li, J.; Hong, K.; Zhang, L on design of an assistive wheelchair system which is directly steered by human thoughts. Int. J. Neural Syst. 2013. In Li et al. [19], The authors evaluated the feasibility of a BCI-based wheelchair that can control the user's mind without additional intervention. In a real-world driving test in a real environment with three healthy participants, the system performed well, suggesting potential applications for people with disabilities in everyday life. To improve ease in the future, the authors considered integrating an infrared sensor and using other types of EEG signals.
5. Carra, M.; Balbinot, A. on Evaluation of a sensorimotor rhythms (SMR) to control a wheelchair. Biosignals and Robotics for a Better and Safer Life (BRC), Rio de Janeiro, Brazil, 18-20. February 2012 [21]. Carra's group described the development of a non-invasive experimental BCI system. The proposed approach used a portable and inexpensive device to generate commands to move a power wheelchair and captured brain signals from the somatosensory cortex without the involvement of peripheral nerves or muscles. Experimental tests conducted in an uncontrolled environment and involving only one volunteer showed promising results and enabled the possibility of future interfacing with real-world situations.
6. G. Reshmi and A. Amal on Design of a BCI System for Piloting a Wheelchair Using Five Class MI Based EEG. In the proceedings of the Third International Conference on Advances in Computing and Communications in Kochi, India on 29–31 August 2013 [17]. To design a BCI

system, Reshmi introduced an EEG-based approach to motor imaging in five classes. The patterns sensed by the sensorimotor cortex are translated into a control signal to control the directional movement of a wheelchair. In fact, the users' movement intentions are classified according to the limb movements, and the pattern recognition results tested by fifty control subjects can be used as a command to move the wheelchair in the specified directions (right, left, forward and backward). and stop doing it.

7. Keun-Tae Kim, Tom Carlson and Seong-Whan Lee on the design of a robotic wheelchair with a motor imagery based Brain Computer interface. In the proceedings of the 2013 International Winter Workshop on Brain Computer Interface (BCI) in Pacific Grove, CA, USA on 3–7 June 2013 [18], Carlson proposed an asynchronous wheelchair system integrated with robotics and computer vision techniques that allows the subject to control the wheelchair by spontaneously performing a motor imagery task. This group introduced the notion of shared control to integrate the user's intelligence with the precise capabilities of a robotic wheelchair in the context of the environment. The authors showed that multiple types of BCI wheelchair users (four healthy subjects, new and experienced) could successfully complete a navigation task. Additionally, the asynchronous MI approach offers users greater flexibility and control over the trajectories actually driven compared to an alternative P300-based system. More specifically, users can spontaneously interact with the wheelchair and voluntarily control the movement at any time, rather than having to wait for external cues. Also, they can dynamically generate intuitive and smooth trajectories instead of relying on predefined routes, reducing idle navigation time.

8. Tom Carlson and Jose del R. Millan on the design of a robotic wheelchair with a motor imagery-based Brain Computer interface. In the proceedings of the 2013 International Winter Workshop on Brain Computer Interface (BCI) in Pacific Grove, CA, USA on 3–7 June 2013 [25]. To overcome a few limitations of various existing solutions, such as gaze dependency and unnecessary stops, Tom Carlson presented an MI-based brain-controlled wheelchair system using an extended five-command protocol. The wheelchair presented could be driven by the user in both gentle and right-angled turns. The system, which was only validated by a healthy subject, could be integrated with various robotic and computer vision sensors via additional channels in the network module to provide appropriate feedback to the user and improve

safety. This approach can enable the user to cope with different environments and reach a goal point with reduced execution time.

9. Sergio Varona-Moya, Francisco Velasco-Álvarez, Salvador Sancha-Ros, Álvaro Fernández-Rodríguez, María J. Blanca, Ricardo Ron-Angevin on Wheelchair navigation with an audio cued, two class motor imagery based brain computer interface system. In Proceedings of the 7th International IEEE/EMBS Conference on Neural Engineering (NER) in Montpellier, France on 22–24 April 2015 [15]. In this research paper, the authors tested the feasibility of driving a customized robotic wheelchair using an MI-based BCI system and the audible cues to inform the subject of the available navigation command at all times. To enable effective and autonomous wheelchair navigation, this group proposed an application interface that provides the user with four navigation commands based on a two-class sensorimotor-rhythm BCI paradigm. The results, obtained from a sample of three healthy Nave participants, suggested that this system appears to be an effective way to autonomously drive a robotic wheelchair and could offer trapped patients a better quality of life.
10. Sim Kok Swee, Kho Desmond Teck Kiang and Lim Zheng You for an EEG Controlled Wheelchair. In the proceedings of the MATEC Web of Conferences in Amsterdam, The Netherlands on 23–25 March 2016 [11], Swee et al. proposed to develop an electric wheelchair that can be controlled directly by the brain and that does not require physical feedback as the user's control input. EEG signals captured with a standard headset are processed and converted into mental commands, and a specially implemented algorithm wirelessly transmits the control signals to the electric wheelchair. The authors expected that this system, tested by five healthy users, could make a new contribution to helping physically disabled people regain their mobility.
11. Zhang, R. Li, Y., Yan, Y. Zhang, H. Wu, S. Yu, T. Gu, Z. on Control of a Wheelchair in an Indoor Environment Based on a Brain–Computer Interface and Automated Navigation. *IEEE Trans. Neural Syst. Rehabil. Eng.* 2015 [13]. Zhang et al. demonstrated the effectiveness of a brain-controlled intelligent wheelchair combining an MI (or alternatively P300)-based BCI and an automated navigation system. Only the MI-based BCI solution was considered for the scope of this review. The proposed wheelchair, tested by three healthy volunteers, has several advantages: (i) it can adapt to changes in the environment; (ii) once the user selects a

destination with the BCI, the system automatically navigates there, reducing the user's workload; (iii) during wheelchair navigation, the user can issue a stop command via the BCI.

12. Ron-Angevin, R.; Velasco-Álvarez, F.; Fernández-Rodríguez, Á.; Díaz-Estrella, A.; Blanca-Mena, M.J.; Vizcaíno-Martín, F.J.. J. Neuroeng. Rehabil. 2017 [9]. Brain-computer interface application: serial auditory interface to control a two-class motor image-based wheelchair. To provide multiple navigation commands without degrading system performance, the study by Ron-Angevin et al. presented a paradigm based on the distinction of only two mental tasks to control the wheelchair, special in that it is embedded with an auditory interface that offers the user four navigational commands. The authors suggested that this system, validated by seventeen healthy participants, could be an effective option to allow potential users with motor neuron disease wheelchair shifting in a controlled environment in the face of more extensive training.
13. Al-Turabi, H.; Al-Junaid, H. on Brain Computer Interface for control of Wheelchair in Smart Environment. In the proceedings of the Smart Cities Symposium 2018 [5], Al-Turabi's group described the experience of developing a complete BCI system capable of instructing a wheelchair to move in different directions using non-invasive EEG brainwaves move. Multiple machine learning algorithms are used to classify human intent to steer the wheelchair and move it in the desired direction. Given the experimental results, performed with only one neurologically healthy volunteer, the authors proposed their system to control other devices and set up a cloud-based system for direct headset-to-wheelchair communication as a future improvement.
14. Yu, Y.; Liu, Y.; Jiang, J.; Yin, E.; Zhou, Z.; Hu, D. An Asynchronous Control Paradigm that is based on Sequential Motor Imagery and It's Application in Wheelchair Navigation. *IEEE Trans. Neural Syst. Rehabil. Eng.* 2018 [7]. In Yu et al. the authors implemented an asynchronous control strategy where the wheelchair commands are generated through a multi-step process based on sequential MI without external prompt information. Although the system was tested on seven healthy volunteers, preliminary test results showed the potential applicability of this navigation strategy in improving the mobility of people with physical disabilities in a real-world environment.

15. Permana, K.; Wijaya, S.K.; Prajitno, P. on a Controlled wheelchair based on brain computer interface using Neurosky Mindwave Mobile Headset *Proc. Int. Symp. Curr. Prog. Math. Sci.* 2019 [3]. Permana's project goal was to control the wheelchair using a motor imagination-based BCI and a wearable EEG device. When conducting a preliminary experimental test using only MI patterns from a single data channel to trigger wheelchair movement, the authors encountered some problems. Classification failed for several wheelchair control signals due to similarities in the EEG patterns related to different motor paradigms. To overcome this limitation, the authors added a new variable (eye movement) as a discriminator of similar data without obtaining interpretable results. In summary, the Permanas system, which has been validated by five ordinary people, still needs further development and improvement.
16. Xiong, M.; Brandenberger, A.; Bulger, M.; Chien, W.; Doyle, A.; Hao, W.; Jiang, J.; Kim, K.; Lahlou, S.; Leung, C.; et al. A Low-Cost, Semi-Autonomous Wheelchair Controlled by Motor Imagery and Jaw Muscle Activation [2]. In *Proceedings of the IEEE International Conference on Systems, Man and Cybernetics (SMC)*, Bari, Italy, 6–9 October 2019. The authors have made several important contributions to the state of the art of BCI. They proposed a wheelchair prototype in which he generated four control commands using hand movement and jaw clenching data collected with a consumer EEG system. In addition, various autonomous driving functions, location trackers and heart rate monitors have been integrated into the system to improve usability and safety. A pilot cohort of seven healthy volunteers was recruited to collect the MI training dataset. Although future experiments and consistent neurofeedback his training procedures were required to validate the prototype, so the system seems closer to the needs and requirements of the real-world context. The authors emphasized that the clinical application of this system is highly dependent on the user's exercise capacity his EMG-Toggle cannot be used for patients with more severe impairments such as CLIS and electrooculography (EOG) proposed other integrative non-muscle signals such as as future work.
17. In this paper, An EEG-based brain-controlled a wheelchair was developed at Brain Computer Interface (BCI) using NeuroSky Mind Wave EEG headset [6]. A quadriplegic patient cannot move any part of the body below the neck. Implementation of this device helps paraplegics to move independently. They proposed a wheelchair which is controlled by variation of attention level of the patient. Turning on or off this device is controlled by double eye blink of the patient The eye blink pattern and alpha and beta wave signals of the patient are

analyzed. One limitation of this work is that the battery of the wheelchair has to be recharged after certain period daily with help of others. This limitation can be overcome by incorporating global positioning system (GPS) based mapping with the wheelchair while the chair will go to a predefined recharging point autonomously when the battery needs to be recharged.

18. Md. Arif Abdulla Samy, Md. Mirazur Rahman, Tafsir Ahmed Khan. In this work, an approach of controlling devices by using Brain Computer Interface (BCI) technology has been demonstrated. Tafsir Ahmed Khan said the research demonstrated an approach to controlling devices using Brain Computer Interface (BCI) technology [10]. The project succeeded by extracting brain signals in response to the subject's body movements and using these signals as some selective commands to control the device. As the signals were extracted by non-invasive electroencephalography, they were processed by a microcontroller and sent as control signals to motors attached to the wheelchair. Extensive analysis of multiple EEG responses was performed throughout the project, including clapping, winking, and hand movements. In addition, the conscious relaxed mind response was also examined. By extracting several selective and well-defined patterns from hand movements, we succeeded in precisely controlling the wheelchair. Despite the infinite number of patterns, only small number of brain patterns were considered, this project demonstrated a very simple and inexpensive approach. Implementing an automation system was not the main purpose of this work. The main concern was to prove that BCI is applicable in practice. This kind of research is necessary to make devices smarter and more compatible with us so that they can be used quickly. This method can be used to drive a wheelchair for a paralyzed person or to communicate with a deaf person.

19. Ming Dong, Fu Lan, Chen Long, Tang Jiabei, Qi Hongzhi, Zhao Xin, Zhou Peng, Zhang Lixin, Jiao Xuejun, Wang Chunhui, Wan Baikun Electric Wheelchair Control System Using Brain-Computer Interface Based on Alpha-Wave Blocking: Brain Computer Interface (BCI) based electric wheelchair control system has been developed [16]. This allows the user to move the wheelchair back and forth, turn left and right without prior learning. This control system uses amplitude enhancement of electroencephalogram (EEG) alpha blockade when his eyes are closed for more than 1 second to form his BCI for switch control of wheelchair movement. The system consisted of BCI control panel, data acquisition, a signal processing unit, and interface control circuitry. In this task, the subject can achieve reliable and stable locomotion

function by using her BCI-based power wheelchair system. compared to ref. realized three-command control of an electric wheelchair by combining blink impulses and attention signals. switch back. Moreover, no prior training is required.

20. Haleema Al-Turabi and Hessa Al-Junaid, The Brain Computer Interface (BCI) offers people with mobility impairments a solution for independent mobility [4]. An intelligent control of wheelchair was proposed. This paper presents a complete hemisphere is consisting of hardware and software components to guide a wheelchair through human intention to move side to side, back and forth and in various directions using non-invasive EEG brain waves. describes his experience was developing a BCI system for Three different machine learning algorithms, K nearest neighbors, support vector machine algorithms, and artificial neural networks are used to classify human intentions moving in the desired direction. The highest accuracy achieved is 79.2% for the support vector machine algorithm. The results show that the system can be used successfully for medical purposes and that the concept can be extended to other applications. Experimental results demonstrate the ability of the proposed BCI system to control other devices and provide future opportunities to improve it. As a future work, it would be good to design a cloud system that communicates with the wheelchair directly from the headset. Additionally, we need to train on more EEG data and test the system on more subjects to improve accuracy.

From the above research papers, we came to know that a substantial amount of work has been done in this area and many researches are still going on. After studying various research papers, we concluded that we can provide a better life for People with Disability (PWD) and we can help them in achieving a better lifestyle.

From the analysis of the research paper we came to conclusion that SSVEP BCI wheelchair system based on SVM(Support Vector Machine) has overcome the limitations of traditional BCI wheelchair that requires less training time, improved accuracy and is less prone to interference.

CHAPTER 3

EXPERIMENTAL SETUP AND METHODOLOGY

3.1 Experimental Setup

- For the study, a total of 35 healthy participants, including 17 women, between the ages of 17 and 34, will be enrolled. There will be a screening for any neurological or visual conditions that can compromise task performance [26].
- The subject's EEG signals will be recorded using a 64-channel EEG device. The reference electrode was placed on the right earlobe and the electrodes were arranged according to the international 10-20 system. At a frequency of 250 Hz, the signal will be amplified and sampled [26].
- Participants will execute a BCI task based on SSVEP in which they must concentrate on characters that are flickering on a screen. With a 0.2 Hz step size, the characters will flicker at frequencies between 8 and 15.8 Hz. Each trial will last 6 seconds, with a visual cue indicating the target stimulus for the first 0.5 seconds of each trial. The participants will be told to resist blinking during the stimulation interval and to transfer their gaze to the target stimulus as rapidly as they can within the cue length [26].
- The EEG data will be pre-processed by applying a bandpass filter between the frequencies of 3 Hz and 50 Hz and segmenting into 6-second epochs that match the stimulus presentation.
- Continuous EEG data was segmented into 6 s periods (500 ms before stimulation and 5.5 s after the start of stimulation). Then reduce the period to 250 Hz. Finally, this data is stored as double-precision floating-point values in MATLAB and labelled as subject indices. For each file, the data loaded into MATLAB results in a 4-dimensional matrix called "data" with dimensions (64, 1500, 40, 6) [26].
- Among the FFTs of all connected channels, the data from Oz&T10 was chosen for further system development, as the strongest SSVEP was observed at Oz. The coefficients of the fundamental and second harmonics of the four target frequencies of the amplitude spectrum were considered as eigenvectors for the classification.
- A linear kernel and a leave-one-out cross-validation (LOOCV) strategy will be used to train the SVM classifier. A grid search approach will be used to fine-tune the regularisation parameter C and the kernel width parameter sigma. The classifier will be trained using the chosen characteristics and the target labels that correspond to them, which stand in for the various flickering characters.

- A new dataset made up of trials from the same individuals will be used to test the trained SVM classifier. Metrics including accuracy, sensitivity, specificity, and the area under the receiver operating characteristic (ROC) curve will be used to evaluate the SVM classifier's performance.
- The SSVEP BCI wheelchair system's movements will be controlled by the SVM classifier's output. To enable motorised wheelchair movements, the output will be sent to a microcontroller unit (MCU) outfitted with a joystick controller. Based on the classifier output and the anticipated movement direction, the movements will be carried out.
- Based on the precision and speed of the wheelchair motions, the SSVEP BCI wheelchair system's performance will be assessed.

3.2 System Overview

The SSVEP BCI wheelchair's algorithmic design, which is based on SVM, goes through several stages. First, data from 35 healthy people who focused on 40 characters flickering at different frequencies was gathered. Six blocks, each having 40 randomised trials, made up the experiment.

A visual cue was shown in each trial, and the individual had to fixate the target stimulus throughout the duration of the cue. The screen then blanked for 0.5 seconds before the next trial started, after which all stimuli flickered on the screen for 5 seconds.

The dataset was used to train and test an SVM classifier that could identify the target stimulus from the subject's SSVEP signals. In order to anticipate the target stimuli in upcoming trials, the classifier was trained using the signals collected throughout the experiment.

The BCI wheelchair system was later upgraded to include the SVM classifier. Eight directional buttons on the wheelchair represented the eight different directions of motion. Based on the predetermined target stimulus, the classifier output decided which direction the wheelchair moved.

In conclusion, compared to conventional BCI wheelchairs, this technology gives users more precise and dependable control over their wheelchair using their SSVEP signals and an SVM classifier.

3.3 Methodology

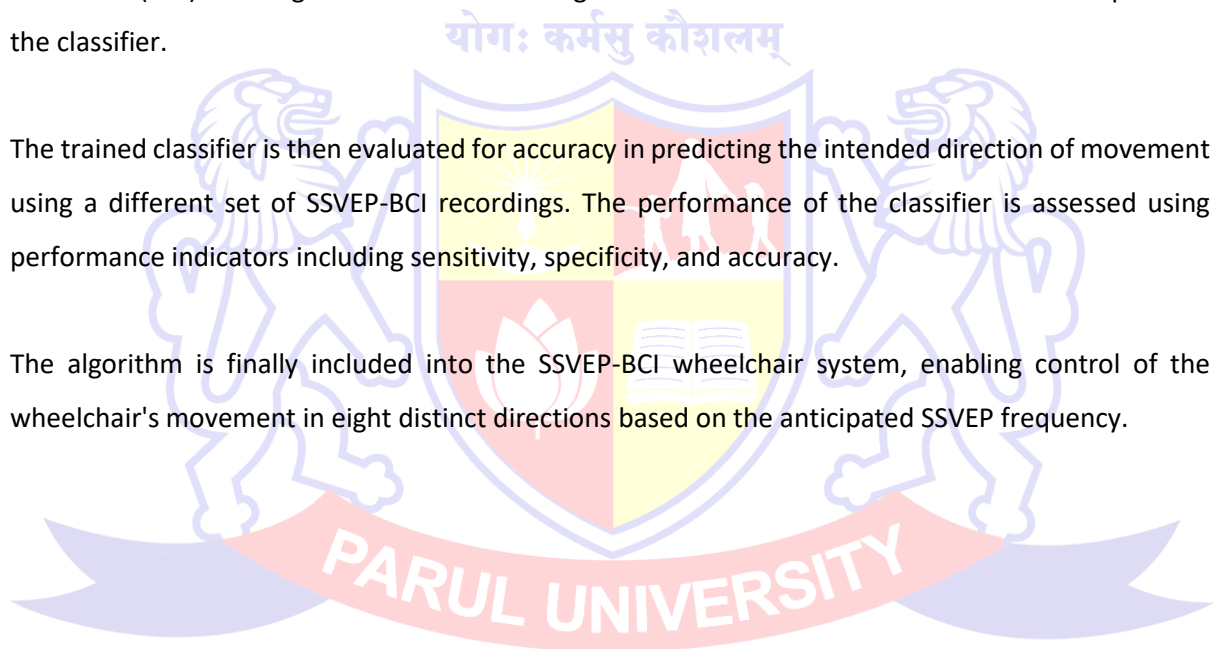
There are several steps in the algorithmic design process for the SSVEP-BCI wheelchair, which uses SVM. First, information from 35 healthy people is gathered through six blocks of 40 randomised trials, each focused on 40 flickering characters at various rates.

The gathered data is then pre-processed to remove any noise or unwanted signals using filtering and artefact removal techniques. Each epoch in the pre-processed dataset corresponds to the time range during which a specific character flicker.

Features are then retrieved from the segmented epochs using methods like the Fast Fourier Transform (FFT) after segmentation. The SVM algorithm is then trained with these features to produce the classifier.

The trained classifier is then evaluated for accuracy in predicting the intended direction of movement using a different set of SSVEP-BCI recordings. The performance of the classifier is assessed using performance indicators including sensitivity, specificity, and accuracy.

The algorithm is finally included into the SSVEP-BCI wheelchair system, enabling control of the wheelchair's movement in eight distinct directions based on the anticipated SSVEP frequency.



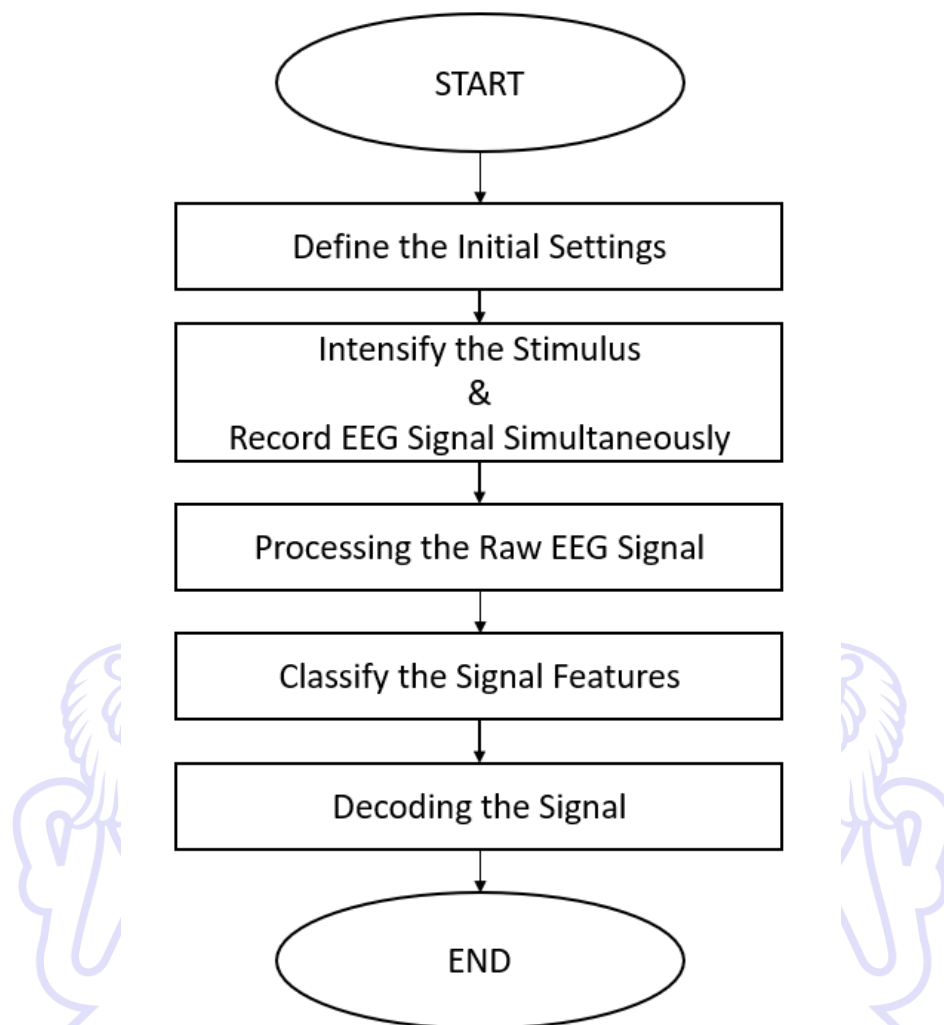


Figure 3.3.1 Flowchart of Process

3.4 Software Description

The numerical computing programme MATLAB makes it simple to develop mathematical and statistical algorithms. Many capabilities for data analysis, visualisation, and modelling are provided by the software.

MATLAB is useful for a wide range of projects, including developing machine learning methods, designing graphical user interfaces, and processing signals and images. The software's syntax is created to be simple to learn and intuitive, allowing users to develop and test their ideas rapidly.

The huge collection of built-in functions and toolboxes in MATLAB is one of its standout qualities. These libraries include tools for a range of uses, including signal processing, optimization, and control

systems. Users can also develop and share their own unique tools, greatly enhancing the software's capabilities.

The MATLAB programming language, which is similar to other programming languages like Python and Java, can be used by users to create scripts or functions for MATLAB. The programme also has a graphical user interface that enables interactive data exploration, algorithm building, and visualisation creation.

MATLAB, which offers a variety of features and capabilities that make it a useful tool for researchers, engineers, and scientists, is a strong tool for numerical calculation and data processing.

3.5 Working of The Project

The frequency range of interest for the SSVEP signal and the noise band is first specified in this code. The signal-to-noise ratio is then initialised in an empty vector, and the pre-processed data for all subjects is initialised in another vector.

The first for loop cycles over each subject's data and computes the SNR by averaging the power in the desired frequency band and the noise band using the Welch method to compute the power spectrum. Results of computing and displaying the SNRs' mean and standard deviation among participants. The EEG signal is bandpass filtered, the DC offset is removed, and the SNR for each subject is calculated in the second for loop, which pre-processes the data for all 35 patients.

1. Define the noise band

The frequency range of the noise in the EEG signal is known as the variable noise band.

2. Initialize empty voters:

The PRE_SNRs and SN vectors are started out as empty 35-length vectors. Each subject's signal-to-noise ratios (SNRs) will be stored in these vectors.

3. Define the frequency band of interest:

The frequency band variable specifies the frequency band of interest for the SSVEP transmission.

4. Compute the power spectrum:

The Welch method and Welch function are used to compute the power spectrum of the EEG data. The estimated power spectral densities that result is kept in PXX, whereas the associated frequencies are kept in f.

5. compute the SNR:

The SNR is calculated as the difference between the noise power and the average power in the frequency band of interest (the average power in the noise band). The presses vector contains the resulting SNR.

6. Compute mean and standard deviation of SNRs:

The mean and standard deviation of the SNRs across all subjects are calculated using the mean and std functions once the SNR for each subject has been determined. The outcomes are kept in separate variables called mean present.

7. Remove DC offset:

By calculating the mean over all time points for each electrode, the Dc offset is determined. The EEG data is then cleaned of this offset.

8. Apply bandpass filter:

The butter function is used to create a bandpass filter, and the filtfilt function is used to apply the bandpass filter to the EEG data. The filtered data that is produced is kept in EEG filtered.

9. Store pre-processed data:

The pre-processed cell array houses the processed EEG data.

10. Compute SNR:

The pre-processed EEG data's SNR is calculated using the same methodology as before. The SNR that results is kept in the SN vector.

The mean and standard deviation of the SNRs across all subjects are calculated using the mean and std functions once the SNR for each subject has been determined.

11. Apply FFT to SSVEP Signal:

FFT is applied to extract the first harmonics and second harmonics of SSVEP signal to form feature matrix and frequency labels.

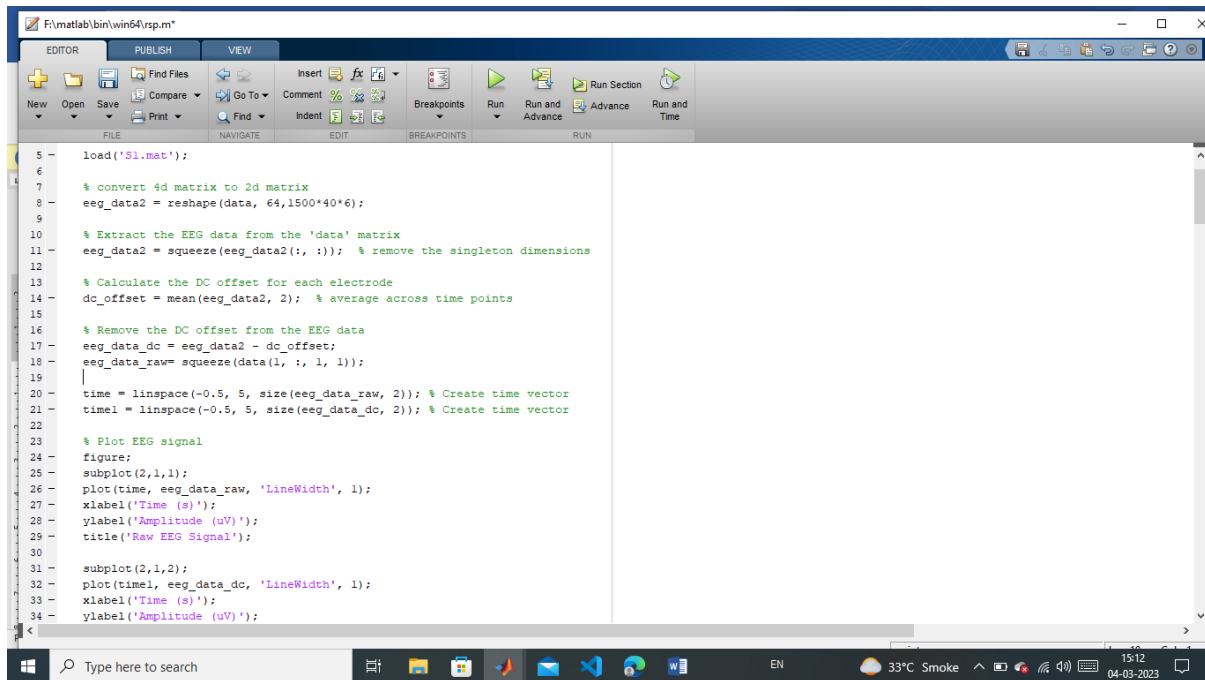
12. Compute SVM of the feature matrix and frequency labels to form the relation between them.



CHAPTER 4

RESULTS

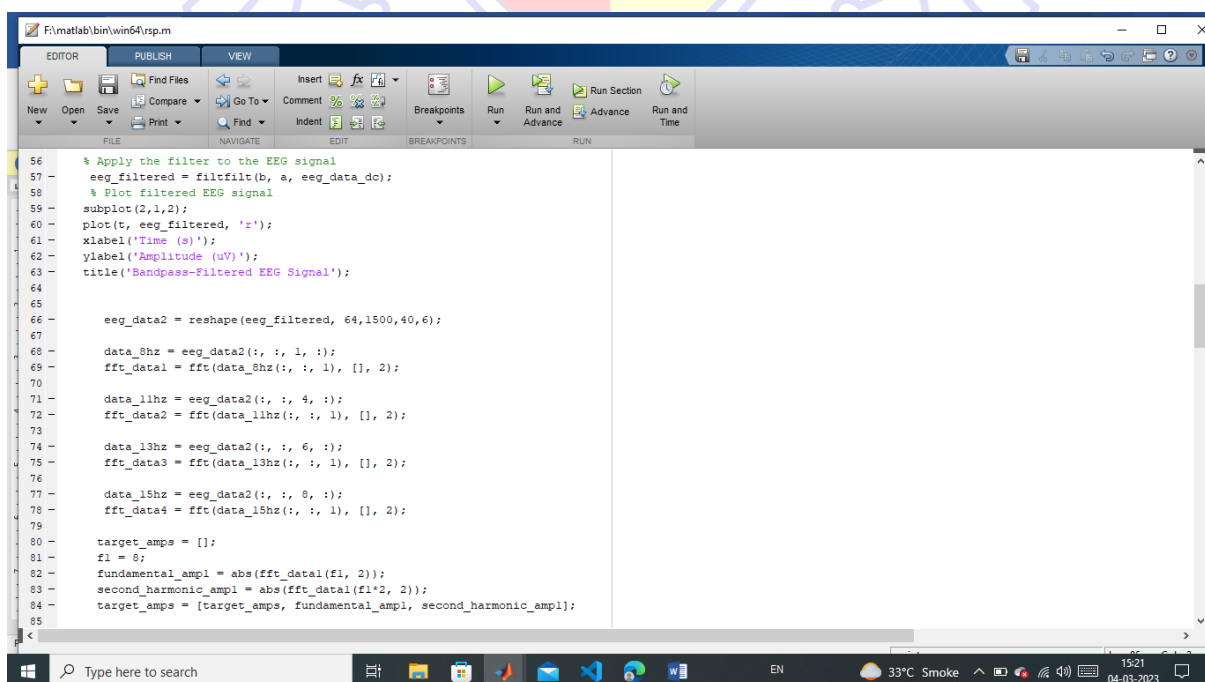
4.1 Results & Stepwise Working



```
5 load('S1.mat');
6
7 % convert 4d matrix to 2d matrix
8 eeg_data2 = reshape(data, 64,1500*40*6);
9
10 % Extract the EEG data from the 'data' matrix
11 eeg_data2 = squeeze(eeg_data2(:, :)); % remove the singleton dimensions
12
13 % Calculate the DC offset for each electrode
14 dc_offset = mean(eeg_data2, 2); % average across time points
15
16 % Remove the DC offset from the EEG data
17 eeg_data_dc = eeg_data2 - dc_offset;
18 eeg_data_raw = squeeze(data(1, :, 1, 1));
19
20 time = linspace(-0.5, 5, size(eeg_data_raw, 2)); % Create time vector
21 time1 = linspace(-0.5, 5, size(eeg_data_dc, 2)); % Create time vector
22
23 % Plot EEG signal
24 figure;
25 subplot(2,1,1);
26 plot(time, eeg_data_raw, 'LineWidth', 1);
27 xlabel('Time (s)');
28 ylabel('Amplitude (uV)');
29 title('Raw EEG Signal');
30
31 subplot(2,1,2);
32 plot(time1, eeg_data_dc, 'LineWidth', 1);
33 xlabel('Time (s)');
34 ylabel('Amplitude (uV)');
```

Figure 4.1.1 Removing DC Offset from The Train Data

This step removes DC-line interference from the EEG signal to improve the SNR of the SSVEP signal.



```
56 % Apply the filter to the EEG signal
57 eeg_filtered = filterfilt(b, a, eeg_data_dc);
58 % Plot filtered EEG signal
59 subplot(2,1,2);
60 plot(t, eeg_filtered, 'r');
61 xlabel('Time (s)');
62 ylabel('Amplitude (uV)');
63 title('Bandpass-Filtered EEG Signal');
64
65
66 eeg_data2 = reshape(eeg_filtered, 64,1500,40,6);
67
68 data_8hz = eeg_data2(:, :, 1, :);
69 fft_data1 = fft(data_8hz(:, :, 1), [], 2);
70
71 data_11hz = eeg_data2(:, :, 4, :);
72 fft_data2 = fft(data_11hz(:, :, 1), [], 2);
73
74 data_13hz = eeg_data2(:, :, 6, :);
75 fft_data3 = fft(data_13hz(:, :, 1), [], 2);
76
77 data_15hz = eeg_data2(:, :, 8, :);
78 fft_data4 = fft(data_15hz(:, :, 1), [], 2);
79
80 target_amps = [];
81 f1 = 8;
82 fundamental_ampl = abs(fft_data1(f1, 2));
83 second_harmonic_ampl = abs(fft_data1(f1*2, 2));
84 target_amps = [target_amps, fundamental_ampl, second_harmonic_ampl];
85
```

Figure 4.1.2 Removing Excess Frequency & Computing FFT From the Train Data

The bandpass filter method extracts the frequency value of interest.

```

101 freq_labels = [1,1,2,2,3,3,4,4]; % where 1 is for moving forward, 2 for moving left, 3 for turning right, and 4 for turning right
102 % repeat the labels for each of the 4 SSVEP frequencies
103 feature_matrix = [1.497611236739828e+03, 3.379487898369906e+02, 1.757579458715490e+03, 3.313254725376159e+03, 3.218872327885150e+03, 9.129347180667744e+02,
104
105
106 load('S3.mat');
107
108 % convert 4d matrix to 2d matrix
109 eeg_data2 = reshape(data, 64,1500*40*6);
110
111 % Extract the EEG data from the 'data' matrix
112 eeg_data2 = squeeze(eeg_data2(:, :)); % remove the singleton dimensions
113
114 % Calculate the DC offset for each electrode
115 dc_offset = mean(eeg_data2, 2); % average across time points
116
117 % Remove the DC offset from the EEG data
118 eeg_data_dc2 = eeg_data2 - dc_offset;
119 eeg_data_raw2 = squeeze(data(1, :, 1, 1)); % Plot the raw and corrected EEG signals for electrode 1, trial 1, target index 1, and block index 1
120
121
122 time = linspace(-0.5, 5, size(eeg_data_raw2, 2)); % Create time vector
123 time1 = linspace(-0.5, 5, size(eeg_data_dc2, 2)); % Create time vector
124
125 % Set filter parameters
126 fs1 = 250; % Sampling rate
127 f_low1 = 3; % Lower cutoff frequency in Hz
128 f_high1 = 50; % Upper cutoff frequency in Hz
129

```

Figure 4.1.3 Removing DC Offset from the Test Data

This step removes DC-line interference from the EEG signal to improve the SNR of the SSVEP signal.

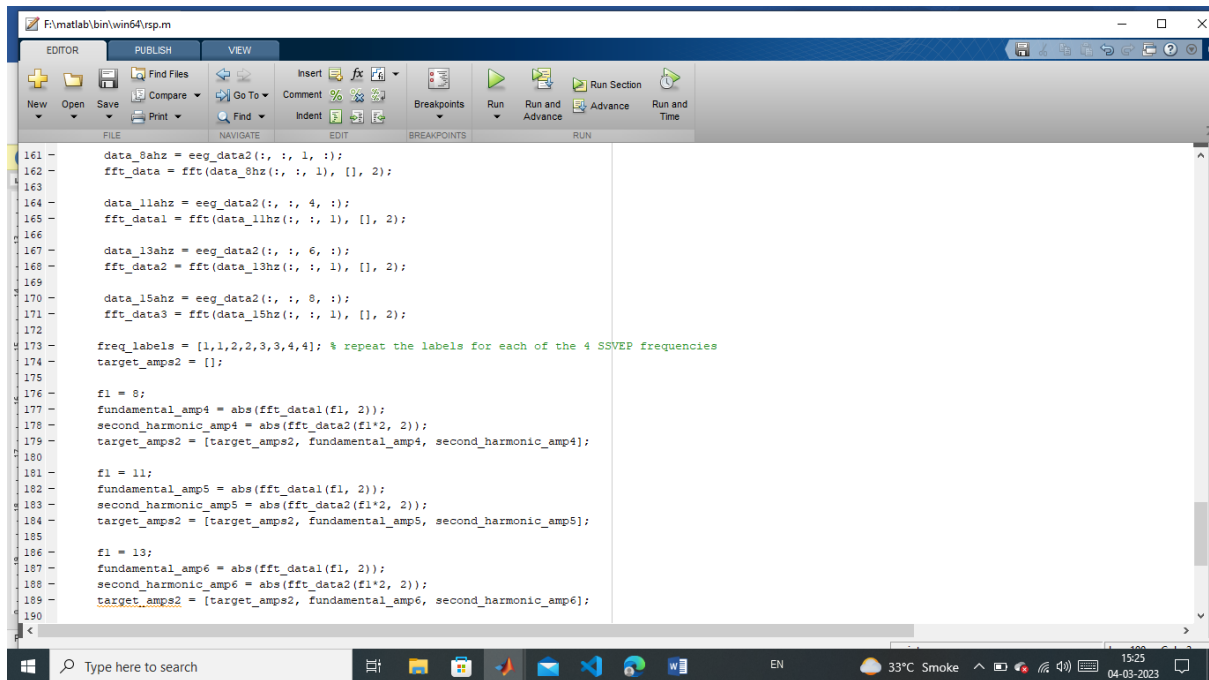
```

130 order1 = 4; % Filter order
131
132 % Design a bandpass filter
133 [b, a] = butter(order1, [f_low1/(fs1/2), f_high1/(fs1/2)], 'bandpass');
134
135 % Apply the filter to the EEG signal
136 eeg_filtered = filtfilt(b, a, eeg_data_dc2);
137 eeg_data2 = reshape(eeg_filtered, 64,1500,40,6);
138 eeg1 = eeg_data2(62,1,6,2);
139 eeg2 = eeg_data2(62,1,4,2);
140
141
142 Fs = 250;
143 N = length(eeg1);
144 f = Fs*(0:N-1)/N;
145 Y = fft(eeg1);
146 figure;
147 plot(f, abs(Y));
148 xlabel('Frequency (Hz)');
149 ylabel('Magnitude');
150 title('FFT of EEG signal recorded ');
151
152 N = length(eeg2);
153 f = Fs*(0:N-1)/N;
154 Z = fft(eeg2);
155 figure;
156 plot(f, abs(Z));
157 xlabel('Frequency (Hz)');
158 ylabel('Magnitude');
159 title('FFT of EEG signal recorded ');
160

```

Figure 4.1.4 Removing Excess Frequency & Computing FFT From the Test Data

The bandpass filter method extract the frequency value of interest.



```

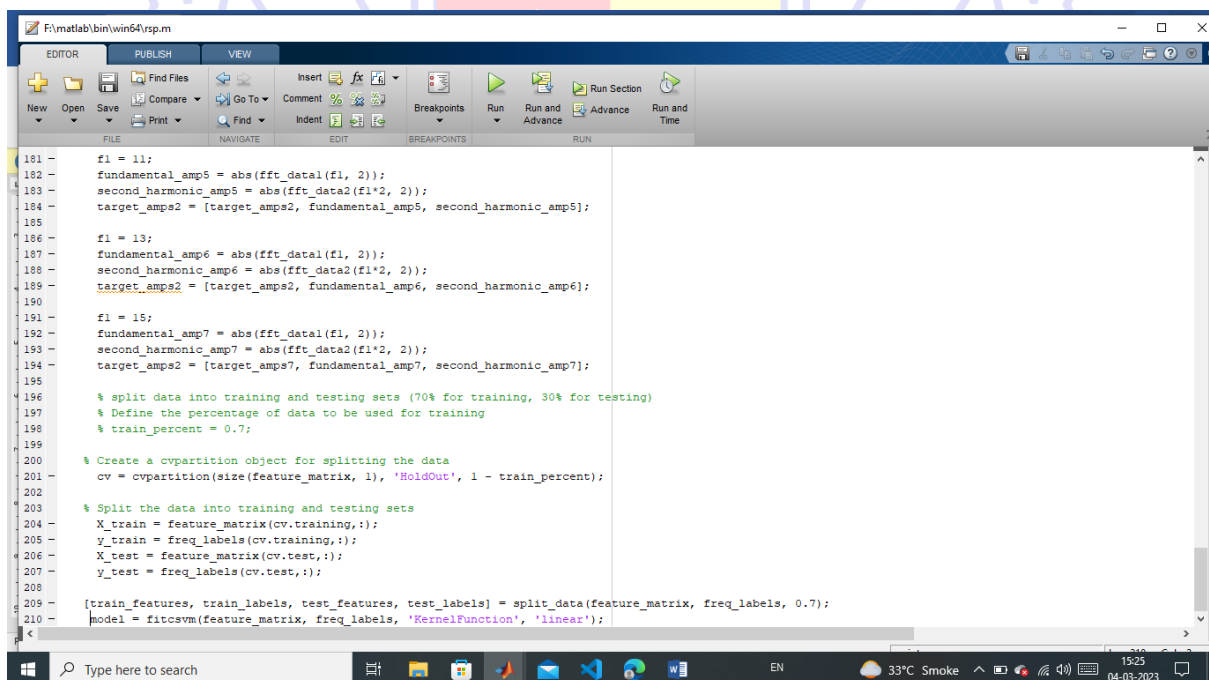
F:\matlab\bin\win64\vsp.m
EDITOR PUBLISH VIEW
+ Find Files
+ Open Save Compare
+ Print
+ Go To
+ Comment
+ Find
+ Indent
+ Breakpoints
+ Run
+ Run and Advance
+ Run Section
+ Run and Time

161 data_8ahz = eeg_data2(:, 1, 1, :);
162 fft_data = fft(data_8hz(:, 1, 1), [], 2);
163
164 data_11ahz = eeg_data2(:, 4, 1, :);
165 fft_data1 = fft(data_11hz(:, 1, 1), [], 2);
166
167 data_13ahz = eeg_data2(:, 6, 1, :);
168 fft_data2 = fft(data_13hz(:, 1, 1), [], 2);
169
170 data_15ahz = eeg_data2(:, 8, 1, :);
171 fft_data3 = fft(data_15hz(:, 1, 1), [], 2);
172
173 freq_labels = [1,1,2,2,3,3,4,4]; % repeat the labels for each of the 4 SSVEP frequencies
174 target_amps2 = [];
175
176 f1 = 8;
177 fundamental_amp4 = abs(fft_data1(f1, 2));
178 second_harmonic_amp4 = abs(fft_data2(f1*2, 2));
179 target_amps2 = [target_amps2, fundamental_amp4, second_harmonic_amp4];
180
181 f1 = 11;
182 fundamental_amp5 = abs(fft_data1(f1, 2));
183 second_harmonic_amp5 = abs(fft_data2(f1*2, 2));
184 target_amps2 = [target_amps2, fundamental_amp5, second_harmonic_amp5];
185
186 f1 = 13;
187 fundamental_amp6 = abs(fft_data1(f1, 2));
188 second_harmonic_amp6 = abs(fft_data2(f1*2, 2));
189 target_amps2 = [target_amps2, fundamental_amp6, second_harmonic_amp6];
190

```

Figure 4.1.5 Computing the 1st And 2nd Harmonics from the EEG Data

These harmonics were used as a feature for training and test dataset for SVM (Support vector machine).



```

F:\matlab\bin\win64\vsp.m
EDITOR PUBLISH VIEW
+ Find Files
+ Open Save Compare
+ Print
+ Go To
+ Comment
+ Find
+ Indent
+ Breakpoints
+ Run
+ Run and Advance
+ Run Section
+ Run and Time

181 f1 = 11;
182 fundamental_amp5 = abs(fft_data1(f1, 2));
183 second_harmonic_amp5 = abs(fft_data2(f1*2, 2));
184 target_amps2 = [target_amps2, fundamental_amp5, second_harmonic_amp5];
185
186 f1 = 13;
187 fundamental_amp6 = abs(fft_data1(f1, 2));
188 second_harmonic_amp6 = abs(fft_data2(f1*2, 2));
189 target_amps2 = [target_amps2, fundamental_amp6, second_harmonic_amp6];
190
191 f1 = 15;
192 fundamental_amp7 = abs(fft_data1(f1, 2));
193 second_harmonic_amp7 = abs(fft_data2(f1*2, 2));
194 target_amps2 = [target_amps2, fundamental_amp7, second_harmonic_amp7];
195
196 % split data into training and testing sets (70% for training, 30% for testing)
197 % Define the percentage of data to be used for training
198 % train_percent = 0.7;
199
200 % Create a cvpartition object for splitting the data
201 cv = cvpartition(size(feature_matrix, 1), 'HoldOut', 1 - train_percent);
202
203 % Split the data into training and testing sets
204 X_train = feature_matrix(cv.training,:);
205 y_train = freq_labels(cv.training,:);
206 X_test = feature_matrix(cv.test,:);
207 y_test = freq_labels(cv.test,:);
208
209 [train_features, train_labels, test_features, test_labels] = split_data(feature_matrix, freq_labels, 0.7);
210 model = fitcsvm(feature_matrix, freq_labels, 'KernelFunction', 'linear');

```

Figure 4.1.6 Using SVM To Compute the Relation Between Feature Matrix & Frequency Labels

4.2 Observation

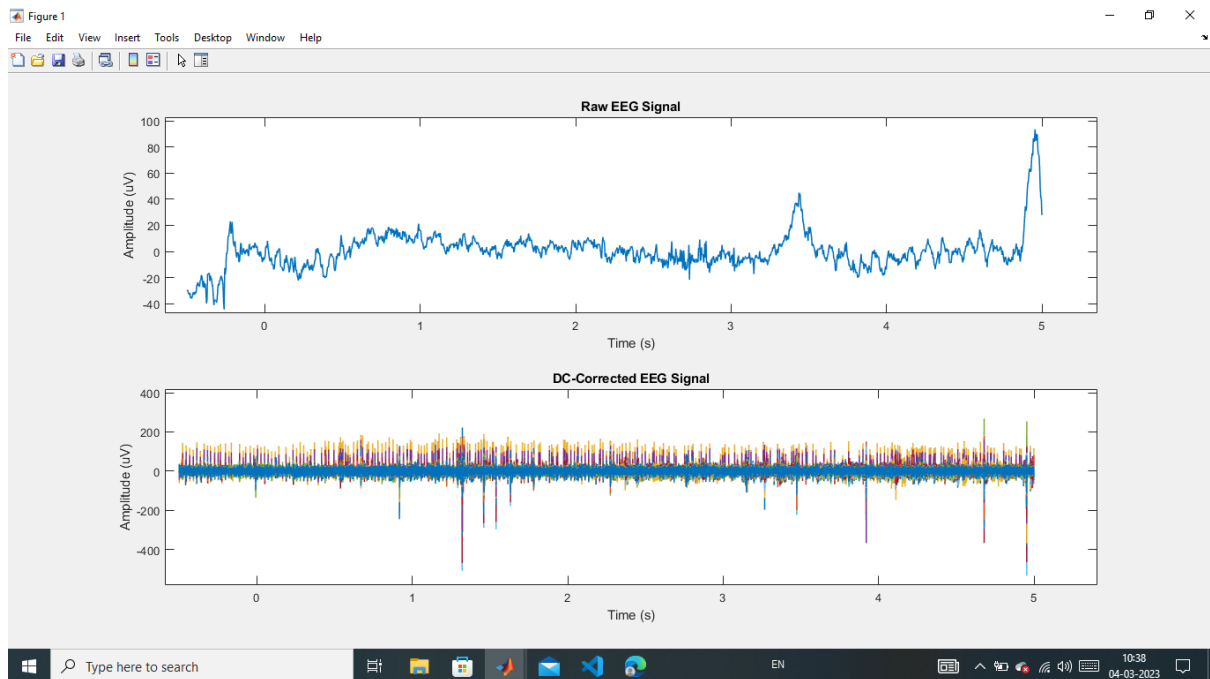


Figure 4.2.1 Raw EEG Signal & DC Corrected EEG Signal

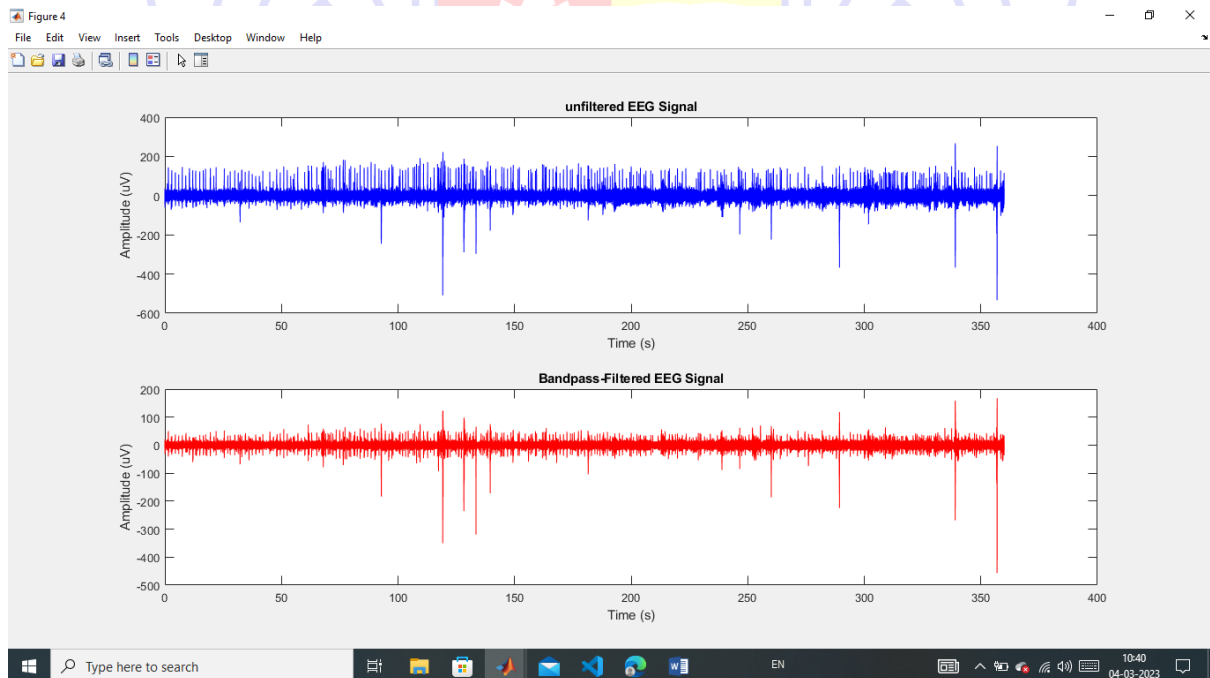


Figure 4.2.2 Unfiltered EEG Signal & Bandpass Filtered EEG Signal

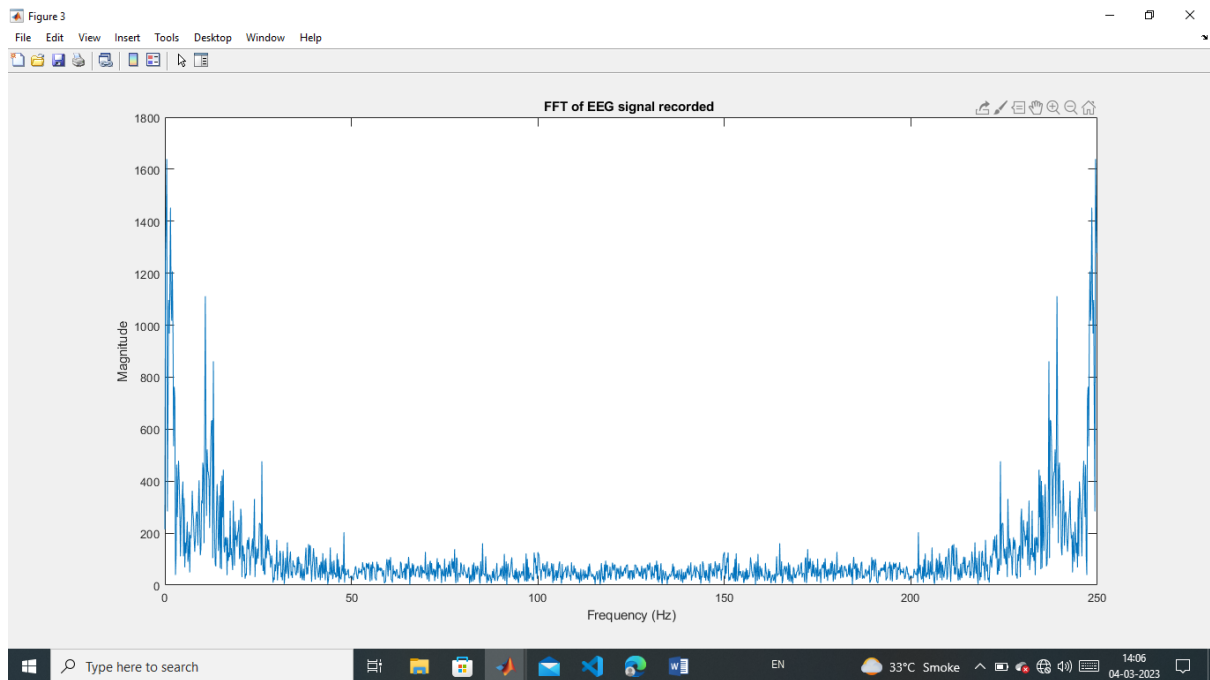


Figure 4.2.3 FFT of EEG Signal Recorded from Oz Electrode

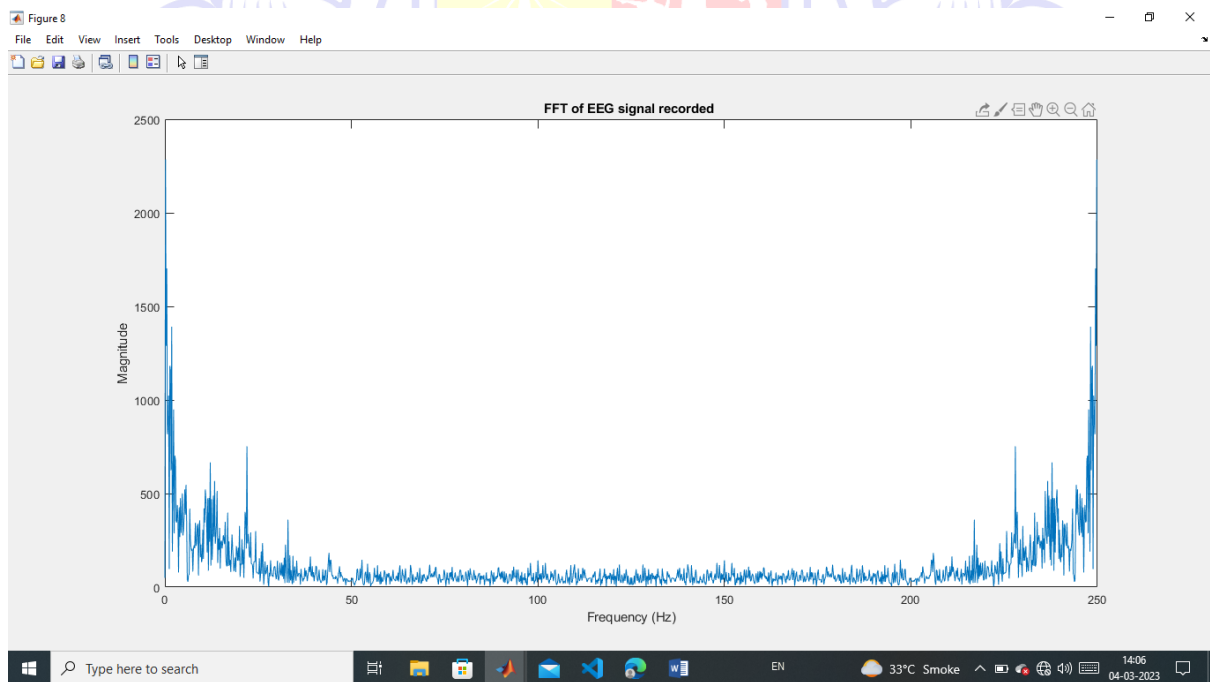


Figure 4.2.4 FFT of EEG Signal Recorded from T10 Electrode

CHAPTER 5

CONCLUSION & FUTURE PERSPECTIVE

5.1 Conclusion

This work presented the development of SSVEP based BCI algorithm using pre-processing method and SSVEP features were extracted using FFT, which is used to control the BCI Wheelchair that navigates in different positions i.e. Forward, Right, Left, and Stop position using SVM (Support Vector Machine) machine learning algorithm.

Fast Fourier Transform (FFT)-based feature extraction method was used to extract the first and second harmonics of SSVEP signal which was used as a feature for SVM (Support Vector machine) algorithm to train on. Pre-processing techniques like removing dc offset from the signal and bandpass filtering increased SNR(Signal-to-noise ratio). The SVM machine learning algorithm exhibited a dependable and effective control approach and overcame the limitations of traditional BCI wheelchair system.

Overall, the SSVEP-based BCI algorithm described in this work represents a promising advancement in the creation of assistive technology for people with mobility disabilities. The algorithm offers a secure and reliable way to regulate how a wheelchair moves by employing non-invasive techniques to find and examine SSVEP signals. As a result, by giving people with disabilities more independence and control over their surroundings, it has the potential to greatly improve their quality of life.

5.2 Future Perspective

The prospective uses of brain-computer interface (BCI) wheelchair technology in healthcare, rehabilitation, and daily life are bright. The following are some potential directions for BCI wheelchair technology development. Enhanced accuracy and dependability: As BCI technology develops, wheelchair control utilising brain signals is anticipated to get better. Better signal processing methods, more complex machine learning algorithms, and improved brain signal collecting technology could be able to accomplish this.

Integration with other technologies: To offer a more complete solution for mobility and freedom for individuals with impairments, BCI wheelchair technology could be integrated with other technologies, like as robots or sensors.

Overall, developments in robotics, machine learning, and neurology are likely to influence the development of BCI wheelchair technology, as well as user needs and healthcare professionals' preferences.

REFERENCES

1. Haoran Zhang ; Enzeng Dong ; Lin Zhu. *Brain-controlled wheelchair system based on SSVEP*. 2020 Chinese Automation Congress. | Publisher IEEE
2. Xiong, M.; Brandenberger, A.; Bulger, M.; Chien, W.; Doyle, A.; Hao, W.; Jiang, J.; Kim, K.; Lahlou, S.; Leung, C.; et al. 2019. *A Low-Cost, Semi-Autonomous Wheelchair Controlled by Motor Imagery and Jaw Muscle Activation*. Italy, 6–9 October 2019. | Publisher IEEE
3. Permana, K.; Wijaya, S.K.; Prajitno, P. 2019 *Controlled wheelchair based on brain computer interface using Neurosky Mindwave Mobile 2*. Proc. Int. Symp. Curr. Prog. Math. Sci. 2019 | Publisher AIP conference proceedings.
4. Haleema Al-Turabi and Hessa Al-Junaied. 2018. *Brain Computer Interface for Wheelchair Control in Smart Environment*. Bahrain, 22-23 April 2018. | Publisher IEEE
5. Al-Turabi, 2018. *Brain Computer Interface for Wheelchair Control in Smart Environment*. In Proceedings of the Smart Cities Symposium 2018, Bahrain, 22–23 April 2018 | Publisher IET digital library
6. Antora Dev, Md. Asifur Rahman, and Nursadul Mamun 2018 *Design of an EEG-based Brain Controlled Wheelchair for Quadriplegic Patients*. 6-9 April 2018. | Publisher IEEE
7. Yu, Y.; Liu, Y.; Jiang, J.; Yin, E.; Zhou, Z.; Hu, D. 2018 *An Asynchronous Control Paradigm Based on Sequential Motor Imagery and Its Application in Wheelchair Navigation*. IEEE Trans. | Publisher IEEE
8. Naijian Chen ; Xu Wng ; Xiuhua Men. 2018. *Hybrid BCI based control strategy of the intelligent wheelchair manipulator system*. 2018 13th IEEE(ICIEA). | Publisher IEEE
9. Ron-Angevin, R.; Velasco-Álvarez, F.; Fernández-Rodríguez, Á.; Díaz-Estrella, A.; Blanca-Mena, M.J.; Vizcaíno-Martín, F.J. 2017 *Brain-Computer Interface application: Auditory serial interface to control a two-class motor-imagery-based wheelchair*. J. Neuroeng. Rehabil. 2017 | Publisher Springer

10. Md. Arif Abdulla Samy, Md. Arif Abdulla Samy, Md. Arif Abdulla Samy. 2016 *Extracting and discriminating selective brain signals in non-invasive manner and using them for controlling a device: a cost-efficient approach to Brain Computer Interface (BCI)*. 22-24 September 2016. | Publisher IEEE
11. Swee, S.K.; Kiang, K.D.T.; You, L.Z. EEG Controlled Wheelchair. 2016. *In Proceedings of the MATEC Web of Conferences*. Amsterdam, The Netherlands, 23–25 March 2016; Volume 51, p. 02011 | Publisher MATEC Web of Conferences.
12. Varona-Moya, S.; Velasco-Alvarez, F.; Sancha-Ros, S.; Fernández-Rodríguez, Á.; Blanca, M.J.; Angevin, R.R. 2015. *Wheelchair navigation with an audio-cued, two-class motor imagery-based brain-computer interface system*. France, 22–24 April 2015 | Publisher IEEE
13. Zhang, R.; Li, Y.; Yan, Y.; Zhang, H.; Wu, S.; Yu, T.; Gu, Z. 2015. *Control of a Wheelchair in an Indoor Environment Based on a Brain–Computer Interface and Automated Navigation*. IEEE Trans. | Publisher IEEE
14. Yanyan Xie ; Xiaou Li *A Brain Controlled Wheelchair based on Common Spatial Pattern*. 2015 (ISBB) | Publisher IEEE
15. Danny Wee-Kiat Ng ; Ying-Wei Soh. *Development of an Automonous BCI Wheelchair*. 2014 IEEE (CIBCI). | Publisher IEEE
16. Ming Dong, Fu Lan, Chen Long, Tang Jiabei, Qi Hongzhi, Zhao Xin, Zhou Peng, Zhang Lixin, Jiao Xuejun, Wang Chunhui, Wan Baikun. 2014. *Electric Wheelchair Control System Using Brain-Computer Interface Based on Alpha-Wave Blocking*. 04 September 2013. | Publisher Springer
17. Reshmi, G.; Amal, 2013. *A Design of a BCI System for Piloting a Wheelchair Using Five Class MI Based EEG*. India, 29–31 August 2013 | Publisher IEEE
18. Kim, K.-T.; Carlson, T.; Lee, S.-W. 2013. *Design of a robotic wheelchair with a motor imagery-based brain-computer interface*. USA, 3–7 June 2013 | Publisher IEEE

19. Li, J.; Liang, J.; Zhao, Q.; Li, J.; Hong, K.; Zhang, 2013. Design of assistive wheelchair system directly steered by human thoughts. *Int. J. Neural Syst.* 2013. | Publisher International Journal of Neural Systems
20. Carlson, T.; Millan, J.D.R. 2013. *Brain-Controlled Wheelchairs: A Robotic Architecture*. IEEE Robot. Autom. Mag. 2013 | Publisher IEEE
21. Carra, M.; Balbinot, 2012. *An Evaluation of sensorimotor rhythms to control a wheelchair*. Brazil, 18–20 February 2012 | Publisher IEEE
22. Carrino, F.; Dumoulin, J.; Mugellini, E.; Khaled, O.A.; Ingold, R. 2012. *A self-paced BCI system to control an electric wheelchair: Evaluation of a commercial, low-cost EEG device*. IEEE 2012 | Publisher IEEE.
23. Tsui, C.S.L.; Gan, J.; Huosheng Hu. 2011. *A Self-Paced Motor Imagery Based Brain-Computer Interface for Robotic Wheelchair Control*. *Clin. EEG Neurosci.* 2011 | Publisher Sage journals
24. Choi, K. 2011. *Control of a vehicle with EEG signals in real-time and system evaluation*. Graefe Arch. Clin. Exp. Ophthalmol. 2011 | Publisher Springer Link.
25. Brice Rebsamen ; Cuntai Guan ; Haihong Zhang. *A Brain Controlled Wheel chair to Navigate in Familiar environments*. 06 December 2010 | Publisher IEEE.
26. <http://bci.med.tsinghua.edu.cn/upload/yijun/Readme.txt>